# Bosonic String Amplitudes From Ribbon Graphs

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**Abstract**

We develop a new method for numerically computing higher-genus CFT correlators and string amplitudes based on Kontsevich's ribbon graph parametrization of the moduli space $\mathcal{M}_{g,n}$ of punctured Riemann surfaces. Using this method, we compute the compact boson partition functions as well as the moduli space integrand for bosonic string vacuum amplitudes at higher genus. Our results pass several checks: the compact boson partition function exhibits T-duality and agrees with known analytic expressions, and the moduli space integrand reproduces the Mumford forms relevant to bosonic string amplitudes.

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# 1 Introduction

# 2 Riemann surfaces from ribbon graphs

## 2.1 Kontsevich's isomorphism

In this section, we review Kontsevich's parametrization of the moduli space $\mathcal{M}_{g,n}$ of genus-$g$, $n$-punctured Riemann surfaces by metric ribbon graphs [1]. Specifically, Kontsevich's isomorphism identifies $\mathcal{M}_{g,n}$ with the combinatorial moduli space $\mathcal{M}_{g,n}^{\mathrm{comb}}$ of metric ribbon graphs:

$$\mathcal{M}_{g,n} \times \mathbb{R}_+^n \xrightarrow{\sim} \mathcal{M}_{g,n}^{\mathrm{comb}}. \tag{2.1}$$

Here, each point of $\mathcal{M}_{g,n}^{\mathrm{comb}}$ corresponds to a metric ribbon graph of genus $g$ with $n$ faces, and the $n$ positive numbers in $\mathbb{R}_+^n$ parametrize the total edge length around each face.

This isomorphism can be constructed explicitly using the Strebel differential, characterized by the following theorem [2, 3].

**Theorem 1** *Let $\Sigma$ be a genus-$g$ Riemann surface with $n$ punctures $\{z_1, \cdots, z_n\}$ that satisfies $n \geq 1$ and $2 - 2g - n < 0$. Given a positive number $p_i \in \mathbb{R}_+$ for each puncture, there exists a unique quadratic differential $\omega = \varphi(z)dz^2$ that satisfies*

1. *$\omega$ is holomorphic on $\Sigma \backslash \{z_1, \cdots, z_n\}$, and $\omega$ has a double pole at each $z_i$.*
2. *The union of all non-compact horizontal trajectories of $\omega$ has measure zero.*
3. *Each compact horizontal trajectory of $\omega$ is a simple loop around one of the punctures $z_i$. Horizontal trajectories $\gamma_i$ around puncture $z_i$ satisfy*

$$\oint_{\gamma_i} \sqrt{\varphi(z)}\, dz = 2\pi p_i. \tag{2.2}$$

It can be proved that each non-compact horizontal trajectory of $\omega$ is critical, namely, that it is bounded by two zeros of $\omega$. The union of all non-compact horizontal trajectories then forms a graph $\Gamma$ with $n$ faces, whose vertices are the zeros of $\omega$ and whose edges are the non-compact horizontal trajectories. The Riemann surface $\Sigma$ can be divided into $n$ domains $D_i$, each swept out by the compact horizontal trajectories around $z_i$, as illustrated in figure 1. This gives a cell decomposition of $\Sigma$: the 2-cells are the domains $D_i$, while the 1-cells and 0-cells are the edges and vertices of the graph $\Gamma$, respectively. If we define the conformally flat metric

$$ds^2 = |\varphi(z)| dz d\bar{z}, \tag{2.3}$$

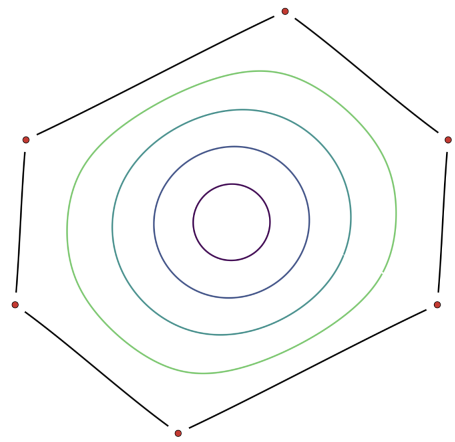

Figure 1: Horizontal trajectories of the Strebel differential $\omega$. The red points correspond to the zeros of $\omega$, the black lines are the non-compact horizontal trajectories, and the colored lines are the compact horizontal trajectories.

then the condition (2.2) tells us that all compact horizontal trajectories around $z_i$ have the same length $2\pi p_i$, and the domains $D_i$ have the geometry of a flat semi-infinite cylinder or, equivalently, a disk. If we define the disk coordinate $w_i$ on the domain $D_i$ by

$$w_i(z) = \exp\left(\frac{i}{p_i}\int^z \sqrt{\omega}\right), \tag{2.4}$$

then the compact horizontal trajectories will have constant radius $|w_i|$, and the puncture $z_i$ will be located at $w_i = 0$. The metric (2.3) will take the form

$$ds^2 = p_i^2 |w_i|^{-2} dw_i d\bar{w}_i. \tag{2.5}$$

The map $\mathcal{M}_{g,n} \times \mathbb{R}_+^n \to \mathcal{M}_{g,n}^{\text{comb}}$ is constructed as follows. Since the edge lengths and the cyclic order of edges at each vertex are preserved under biholomorphism, the graph $\Gamma$ carries the structure of a metric ribbon graph. The map is then defined by sending $(\Sigma, \{p_i\})$ to the metric ribbon graph $\Gamma$ constructed in this way. Conversely, from a metric ribbon graph $\Gamma$, the Riemann surface $\Sigma$ can be reconstructed by gluing a semi-infinite flat cylinder along each face of $\Gamma$, where the circumference of the cylinder is equal to the total edge length around that face, as shown in figure 2. This process defines the inverse map $\mathcal{M}_{g,n}^{\text{comb}} \to \mathcal{M}_{g,n} \times \mathbb{R}_+^n$.

The combinatorial moduli space $\mathcal{M}_{g,n}^{\text{comb}}$ comes with a natural cell decomposition, in which each cell is labeled by the topology, or combinatorial type, of the ribbon graph. In particular, the top-dimensional cells correspond to trivalent graphs, namely, graphs with only cubic vertices. In this case, each zero of the Strebel differential has degree 1. The full combinatorial moduli space $\mathcal{M}_{g,n}^{\text{comb}}$ is obtained by gluing these cells together along boundaries where one or more

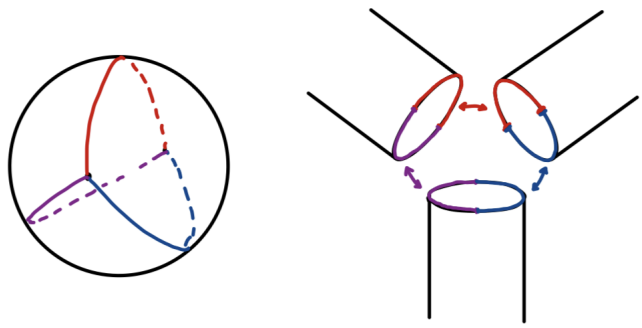

Figure 2: Reconstructing a 3-punctured sphere by gluing 3 semi-infinite cylinders. The left picture shows the ribbon graph $\Gamma$.

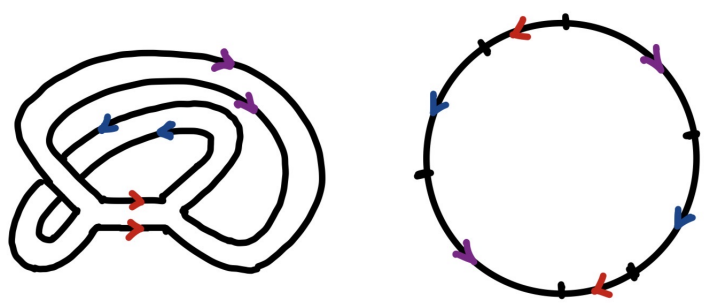

Figure 3: Reconstructing a 1-punctured torus by gluing boundary segments of the unit disk. Note that the boundary segments are glued in the opposite orientation.

edge lengths degenerate to zero. For a trivalent graph with $n$ faces, its dual graph defines an $n$-vertex triangulation of the Riemann surface $\Sigma$, and the number of top-dimensional cells in $\mathcal{M}_{g,n}^{\text{comb}}$ can be counted by enumerating $n$-vertex triangulations on $\Sigma$.

In the rest of this paper, we focus on the case of Riemann surfaces with a single puncture. In this case, the Riemann surface $\Sigma$ can be reconstructed from the ribbon graph by identifying segments on the boundary of a unit disk $D$, with the puncture located at the origin of $D$. The example of reconstructing a 1-punctured torus by gluing boundary segments of $D$ is illustrated in figure 3.

Since the top-dimensional cells in $\mathcal{M}_{g,1}^{\text{comb}}$ correspond to trivalent graphs, the ribbon graph at a generic point in the moduli space has only cubic vertices. The number of vertices $V$ and edges $E$ in the ribbon graph $\Gamma$ therefore satisfy $3V = 2E$. Together with Euler's formula $V - E + F = 2 - 2g$, one can find

$$E = 6g - 3, \quad V = 4g - 2. \tag{2.6}$$

| Genus | Number of edges | Number of vertices | Number of ribbon graph topologies |
|---|---|---|---|
| 1 | 3 | 2 | 1 |
| 2 | 9 | 6 | 9 |
| 3 | 15 | 10 | 1726 |
| 4 | 21 | 14 | 1349005 |
| 5 | 27 | 18 | 2169056374 |

Table 1: Number of edges, vertices, and topologies of genus-$g$ ribbon graphs with a single face.

Thus, reconstructing $\Sigma$ requires gluing $2E = 6(2g-1)$ segments along the boundary of $D$, and different ways of gluing correspond to different topologies of the ribbon graph. The number of one-vertex triangulations of a genus-$g$ surface was determined by Bacher and Vdovina [4], and the corresponding numbers of ribbon graph topologies are listed in table 1.

## 2.2 Holomorphic data from ribbon graphs: Genus 1

Given a metric ribbon graph $\Gamma$, one can numerically compute a basis of holomorphic 1-forms on $\Sigma$. For simplicity, we first present the algorithm for computing the holomorphic 1-form $f(z)dz$ on a genus-1 surface.

The 1-punctured torus can be reconstructed by gluing 3 pairs of boundary segments on the unit disk. For genus 1, there is only one ribbon-graph topology, and the corresponding gluing map is the same as the one illustrated in figure 3. If we parametrize the unit circle by the angular coordinate $\sigma \in (0, 2\pi)$, the gluing map reads

$$\begin{aligned} &\sigma \sim \pi + \theta_1 - \sigma, \quad 0 < \sigma < \theta_1, \\ &\sigma \sim \pi + \theta_1 + \theta_2 - \sigma, \quad \theta_1 < \sigma < \theta_2, \\ &\sigma \sim 2\pi + \theta_2 - \sigma, \quad \theta_2 < \sigma < \pi. \end{aligned} \tag{2.7}$$

The angles $\theta_1, \theta_2 - \theta_1$, and $\pi - \theta_2$ are the angular sizes of the three segments. In other words, they are the edge lengths of the ribbon graph if we take the radius $p$ to be 1. The moduli space $\mathcal{M}_{1,1}$ can be parametrized by the two angles $\theta_1$ and $\theta_2$.

Let us denote the disk coordinate by $z$, where the unit circle is given by $|z| = 1$. The holomorphic 1-form $f(z)dz$ should be regular for $|z| < 1$, and it should diverge at the Strebel vertices, namely, at the zeros $z = \pm 1$, $\pm e^{i\theta_1}$, and $\pm e^{i\theta_2}$ of the Strebel differential $\omega = \varphi(z)dz^2$. In the flat torus coordinate $u$, where the holomorphic 1-form is simply $du$, the disk coordinate $z$ can be related to $u$ by equation (2.4), which implies

$$f^{-1}(z) = \frac{du}{dz} = iz\sqrt{\varphi(u)}. \tag{2.8}$$

Since the ribbon graph $\Gamma$ has only cubic vertices, each zero $u_i$ of $\omega$ is of degree one, *i.e.*, $\varphi(u) \sim (u - u_i)$ near $u_i$, and hence $f(z)$ diverges like $(u - u_i)^{-1/2}$. Translating back to the disk coordinate, one finds the following divergent behavior of $f(z)$ around the zeros $z_i$:

$$f(z) \sim (z - z_i)^{-1/3}. \tag{2.9}$$

Based on this divergent behavior, one can make the following ansatz:

$$f(z)dz = \frac{\sum_{n=1}^{\infty} a_n z^{n-1}}{(1-z^2)^{\frac{1}{3}}(1-z^2e^{-2i\theta_1})^{\frac{1}{3}}(1-z^2e^{-2i\theta_2})^{\frac{1}{3}}}dz. \tag{2.10}$$

The coefficients $a_n$ can be determined numerically as follows. We discretize the unit circle into $L$ points, where $L$ is an even number. The discretized points $z(k)$ are located at

$$z(k) = e^{\frac{2\pi i}{L}(k-\frac{1}{2})}, \quad k = 1, \cdots, L, \tag{2.11}$$

and we make the identifications

$$\begin{aligned} &z(k) \sim z(\tfrac{L}{2} + \ell_1 + 1 - k), \quad 1 \le k \le \ell_1, \\ &z(k) \sim z(\tfrac{L}{2} + 2\ell_1 + \ell_2 + 1 - k), \quad \ell_1 \le k \le \ell_1 + \ell_2, \\ &z(k) \sim z(L + \ell_1 + \ell_2 + 1 - k), \quad \ell_1 + \ell_2 \le k \le \tfrac{L}{2}, \end{aligned} \tag{2.12}$$

as a discretized version of the gluing map (2.7). We will denote the point identified with $z(k)$ by $z(k')$. The relative segment lengths $\frac{\ell_1}{L}, \frac{\ell_2}{L}$ can be identified with the moduli $\theta_1, \theta_2 - \theta_1$, respectively. The Strebel vertices $\pm 1, \pm e^{i\theta_1}, \pm e^{i\theta_2}$ are located halfway between the discretized points at the edges of identified sections of the boundary.

In order to solve (2.10), we can truncate the Fourier coefficients $a_n$ by restricting to the lowest $\frac{L}{2}$ modes, normalize with $a_1 = 1$, and solve for $a_2, \cdots, a_{L/2}$ by imposing the condition that $f(z)dz$ matches across identified discretized points:

$$\begin{aligned} &\frac{\sum_{n=1}^{\frac{L}{2}} a_n e^{in(2\pi\frac{k-\frac{1}{2}}{L})}}{(1-e^{4i\pi\frac{k-\frac{1}{2}}{L}})^{\frac{1}{3}}(1-e^{4i\pi\frac{k-\frac{1}{2}-\ell_1}{L}})^{\frac{1}{3}}(1-e^{4i\pi\frac{k-\frac{1}{2}-\ell_1-\ell_2}{L}})^{\frac{1}{3}}} \\ &\quad = -\frac{\sum_{n=1}^{\frac{L}{2}} a_n e^{in(2\pi\frac{k'-\frac{1}{2}}{L})}}{(1-e^{4i\pi\frac{k'-\frac{1}{2}}{L}})^{\frac{1}{3}}(1-e^{4i\pi\frac{k'-\frac{1}{2}-\ell_1}{L}})^{\frac{1}{3}}(1-e^{4i\pi\frac{k'-\frac{1}{2}-\ell_1-\ell_2}{L}})^{\frac{1}{3}}}, \qquad 1 \le k \le \tfrac{L}{2}, \end{aligned} \tag{2.13}$$

where the relative sign is due to flipping the orientation of $dz$.

Once we have solved for the holomorphic 1-form, we can numerically compute the usual complex modulus $\tau$ of the torus by integrating it against a symplectic basis $\alpha, \beta$ of 1-cycles

$$\tau = \frac{\int_\beta f(z)dz}{\int_\alpha f(z)dz}. \tag{2.14}$$

In the disk frame, both of these cycles are given by chords that connect the Strebel vertices, as illustrated in figure 4.

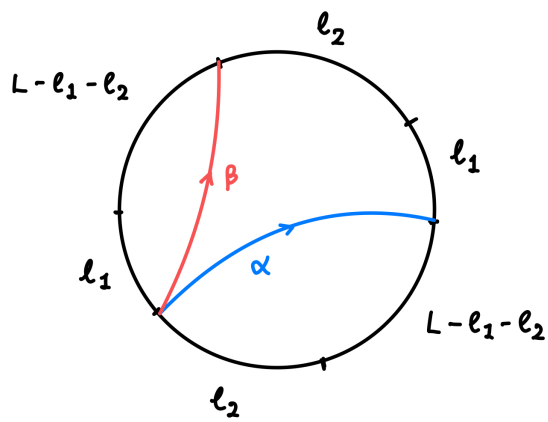


Figure 4: Symplectic basis of 1-cycles on the torus.

## 2.3 Holomorphic data from ribbon graphs: Higher genus

The algorithm above generalizes to the case $g \geq 2$. The genus-$g$ Riemann surface can be reconstructed by gluing $12g-6$ boundary segments on the disk. Upon discretizing the unit circle into $L$ sites, the moduli space $\mathcal{M}_{g,1}$ can be parametrized by the $6g-4$ independent edge lengths.

Going counterclockwise from $z=1$, we label the boundary segments $S_i$ from $i=1$ to $i=12g-6$. We label the edges by $E_a$ with $a=1,\cdots,6g-3$, where each edge $E_a$ pairs two segments $S_{i_1(a)}$ and $S_{i_2(a)}$ and glues them with opposite orientation. The number of discretized sites contained in $E_a$ is denoted by $\ell_a$. We always take $i_1(a)<i_2(a)$. Conversely, we denote the edge associated with segment $i$ by $a(i)$.

The map $a(i)$ depends on the ribbon-graph topology. Given a ribbon-graph topology, the corresponding top-dimensional cell in $\mathcal{M}_{g,1}^{\text{comb}}$ can be parametrized by the relative edge lengths $\frac{\ell_a}{L}$. In this language, the gluing map on the unit circle can be written as

$$z(k) \sim z\left(\sum_{j\leq i_1(a)} \ell_{a(j)} + \sum_{k<i_2(a)} \ell_{a(k)} - k + 1\right), \quad \sum_{j<i_1(a)} \ell_{a(j)} + 1 \leq k \leq \sum_{j\leq i_1(a)} \ell_{a(j)}, \tag{2.15}$$

where the discretized points $z(k)$ are given by (2.11). The Strebel vertices are located at

$$z_i = \exp\left(\frac{2\pi i}{L}\sum_{j<i} \ell_{a(j)}\right). \tag{2.16}$$

One additional complication is that, for higher genus, the independent boundary points are not simply $z(k)$ with $1\leq k\leq L/2$, since the segments in the same half-circle can be glued together. We denote the set of independent variables by $\mathcal{K} = \cup_{a=1}^{6g-3}\{k, k\in S_{i_1(a)}\}$.

On a genus-$g$ surface, there are $g$ linearly independent holomorphic 1-forms. We start with the following ansatz for each holomorphic 1-form

$$f(z)dz = \frac{\sum_{n=1}^{\infty} a_n z^{n-1}}{\prod_{i=1}^{12g-6}\left(1 - z\exp\left(-\frac{2\pi i}{L}\sum_{j<i}\ell_{a(j)}\right)\right)^{\frac{1}{3}}}dz, \tag{2.17}$$

and solve for the coefficients $a_1, \cdots, a_{L/2}$ by matching the values of $f(z)dz$ on the identified boundary points. This produces a system of linear equations for the coefficients $a_n$ of the form

$$\begin{aligned}\frac{\sum_{n=1}^{\frac{L}{2}} a_n z(k)^{n-1}}{\prod_{i=1}^{12g-6}\left(1 - z(k)\exp\left(-\frac{2\pi i}{L}\sum_{j<i}\ell_{a(j)}\right)\right)^{\frac{1}{3}}} \\ = -\frac{\sum_{n=1}^{\frac{L}{2}} a_n z(k')^{n-1}}{\prod_{i=1}^{12g-6}\left(1 - z(k')\exp\left(-\frac{2\pi i}{L}\sum_{j<i}\ell_{a(j)}\right)\right)^{\frac{1}{3}}}, \quad k \in \mathcal{K}.\end{aligned} \tag{2.18}$$

This system of linear equations can be written in the form $Ba = 0$, where $a = (a_1, \cdots, a_{L/2}) \in \mathbb{C}^{L/2}$ is a vector in the coefficient space, and $B$ is a $\frac{L}{2} \times \frac{L}{2}$ matrix defined by

$$\begin{aligned}B_{k,n} = \frac{z(k)^{n-1}}{\prod_{i=1}^{12g-6}\left(1 - z(k)\exp\left(-\frac{2\pi i}{L}\sum_{j<i}\ell_{a(j)}\right)\right)^{\frac{1}{3}}} \\ + \frac{z(k')^{n-1}}{\prod_{i=1}^{12g-6}\left(1 - z(k')\exp\left(-\frac{2\pi i}{L}\sum_{j<i}\ell_{a(j)}\right)\right)^{\frac{1}{3}}}.\end{aligned} \tag{2.19}$$

Since there are $g$ linearly independent holomorphic 1-forms, the matrix $B$ has a $g$-dimensional kernel. To obtain a numerically stable basis of holomorphic 1-forms, we first find the $g$ singular vectors $v_I$ of $B$ with the smallest singular values, and choose $g$ coefficients $a_{n_1}, \cdots, a_{n_g}$ such that the matrix $H_{IJ} = (v_I)_{n_J}$ has the largest possible smallest singular value. We then normalize the $J$-th holomorphic 1-form $\omega_J = f^{(J)}(z)dz$ by imposing $a_{n_I}^{(J)} = \delta_{IJ}$ for $I = 1, \cdots, g$. The remaining coefficients $a_n^{(J)}$ are then obtained by a Hermitian block solve.

For each edge $E_a$, we define a chord connecting the midpoints of $S_{i_1(a)}$ and $S_{i_2(a)}$. One can then choose a symplectic basis $\{\alpha^I, \beta_I\}_{I=1}^g$ of 1-cycles among these chords, with intersection numbers satisfying

$$\alpha^I \cdot \alpha^J = \beta_I \cdot \beta_J = 0, \quad \alpha^I \cdot \beta_J = \delta^I_J. \tag{2.20}$$

The period matrix $\Omega$ of the Riemann surface can then be computed as

$$\Omega = \mathcal{A}^{-1}\mathcal{B}, \quad \mathcal{A}_I{}^J = \oint_{\alpha^J} \omega_I, \quad \mathcal{B}_{IJ} = \oint_{\beta_J} \omega_I. \tag{2.21}$$

Further geometric quantities on the Riemann surface, such as the Riemann constant vector and the prime form, can be computed using the holomorphic 1-forms and the period matrix. These quantities are used in the computation of $bc$ ghost correlators; the details are presented in section 4.

# 3 Free boson CFT from ribbon graphs

## 3.1 The free boson partition function

The partition function of the free boson CFT on a Riemann surface of arbitrary genus can be computed in the ribbon-graph formalism by performing the path integral on the reconstructed surface. For simplicity, we again consider the genus-1 case first. On a torus parametrized by the angle moduli $(\theta_1, \theta_2)$, the free boson partition function can be written as a path integral over the disk, with delta functions implementing the gluing of boundary segments:

$$\begin{aligned} Z(\theta_1,\theta_2) = \int [\mathcal{D}X] \exp(-S[X]) \prod_{0<\sigma<\theta_1} \delta(X(1,\sigma) - X(1,\pi+\theta_1-\sigma)) \\ \times \prod_{\theta_1<\sigma<\theta_2} \delta(X(1,\sigma) - X(1,\pi+\theta_1+\theta_2-\sigma)) \prod_{\theta_2<\sigma<\pi} \delta(X(1,\sigma) - X(1,2\pi+\theta_2-\sigma)), \end{aligned} \tag{3.1}$$

where we have parametrized the boson field $X(r,\sigma)$ in polar coordinates $z = re^{i\sigma}$. Our convention for the free boson action $S[X]$ is such that

$$S[X] = \frac{1}{2\pi\alpha'} \int_\Sigma d^2z \; \partial X \bar{\partial} X. \tag{3.2}$$

We also know that the path integral inside the disk is given by the wavefunctional $\Psi_1[X(\sigma)]$ of the identity operator

$$\Psi_1[X(\sigma)] \propto \exp\left(-\frac{1}{\alpha'} \sum_{m\geq 1} m X_m X_{-m}\right), \tag{3.3}$$

where $X_m$ is the $m$-th Fourier mode of $X(\sigma)$. Therefore,

$$\begin{aligned} Z(\theta_1,\theta_2) = \int [DX(\sigma)]_{0<\sigma<2\pi} \Psi_1[X(\sigma)] \prod_{0<\sigma<\theta_1} \delta(X(\sigma) - X(\pi+\theta_1-\sigma)) \\ \times \prod_{\theta_1<\sigma<\theta_2} \delta(X(\sigma) - X(\pi+\theta_1+\theta_2-\sigma)) \prod_{\theta_2<\sigma<\pi} \delta(X(\sigma) - X(2\pi+\theta_2-\sigma)). \end{aligned} \tag{3.4}$$

Since the identity wavefunctional is Gaussian in $X$, the torus partition function $Z(\theta_1, \theta_2)$ can be computed by a Gaussian integral. This relation generalizes straightforwardly to higher genus and to operators other than the identity, and the final integral remains Gaussian.

To compute the partition function numerically, we again discretize the unit circle into $L$ sites, with a variable $X(k)$ on each site $k = 1, \cdots, L$. The wavefunctional (3.3) can be approximated by

$$\begin{aligned}\Psi_1[X] &\propto \exp\left[-\frac{1}{\pi\alpha' L}\sum_{m=1}^{L/2}\sin\frac{\pi m}{L}\sum_{k,\ell=1}^{L}X(k)X(\ell)e^{2\pi i\frac{m}{L}(k-\ell)}\right] \\ &= \exp\left[-\frac{1}{2\pi\alpha' L}\sum_{k,r=1}^{L}X(k)X(k+r)\frac{\sin\frac{\pi}{L}}{\cos\frac{2\pi r}{L}-\cos\frac{\pi}{L}}\right].\end{aligned} \tag{3.5}$$

Note that we have summed only over the modes with $m$ from 1 to $L/2$. This is because there are only $L$ independent discrete Fourier modes on a circle with $L$ sites, so we must restrict to the modes with $1 \leq m \leq L/2$ in order to write down $\sum X_m X_{-m}$. The delta functions on the boundary can be imposed by making the identifications

$$\begin{aligned}&X(\tfrac{L}{2}+\ell_1+1-k) = X(k), \quad 1 \leq k \leq \ell_1, \\ &X(\tfrac{L}{2}+2\ell_1+\ell_2+1-k) = X(k), \quad \ell_1+1 \leq k \leq \ell_1+\ell_2, \\ &X(L+\ell_1+\ell_2+1-k) = X(k), \quad \ell_1+\ell_2+1 \leq k \leq \tfrac{L}{2}.\end{aligned} \tag{3.6}$$

The partition function (3.4) is then discretized into

$$Z(L;\ell_1,\ell_2) = \int\prod_{k=1}^{L}dX(k)\prod_{k=1}^{L/2}\delta(X(k)-X(k'))\,\Psi_1[X]. \tag{3.7}$$

After replacing each $X(k)$ with $\frac{L}{2}+1 \leq k \leq L$ by its identified counterpart in (3.5), the wavefunctional $\Psi_1$ can be written as

$$\Psi_1[X] \propto \exp\left[-\frac{1}{\pi}\sum_{k,\ell=1}^{\frac{L}{2}}\mathbf{A}_{k\ell}(L;\ell_1,\ell_2)X(k)X(\ell)\right], \tag{3.8}$$

where the matrix $\mathbf{A}$ is given by

$$\begin{aligned}\mathbf{A}_{k\ell}(L;\ell_1,\ell_2) = \frac{1}{2\alpha' L}\Bigg[&\frac{\sin\frac{\pi}{L}}{\cos\frac{2\pi(k-\ell)}{L}-\cos\frac{\pi}{L}}+\frac{\sin\frac{\pi}{L}}{\cos\frac{2\pi(k-\ell')}{L}-\cos\frac{\pi}{L}} \\ &+\frac{\sin\frac{\pi}{L}}{\cos\frac{2\pi(k'-\ell)}{L}-\cos\frac{\pi}{L}}+\frac{\sin\frac{\pi}{L}}{\cos\frac{2\pi(k'-\ell')}{L}-\cos\frac{\pi}{L}}\Bigg]\end{aligned} \tag{3.9}$$

and $k'$ is the partner site of $k$ determined by the identifications (3.6).

The eigenvalues of the matrix $\mathbf{A}_{k\ell}$ are all positive except for a single zero. This zero mode corresponds to the constant mode of $X(\sigma)$ on the unit circle. The integration over $X(k)$, $1 \leq k \leq \frac{L}{2}$, then produces, up to a moduli-independent constant normalization factor, the determinant

$$Z(L;\ell_1,\ell_2) \propto \frac{1}{\sqrt{\det \mathbf{A}'(L;\ell_1,\ell_2)}}, \tag{3.10}$$

where the prime on $\mathbf{A}$ denotes the removal of the zero mode $\sum_k X(k)$. This can be done by imposing the condition

$$X(\tfrac{L}{2}) = -\sum_{k=1}^{\frac{L}{2}-1} X(k), \tag{3.11}$$

and the $(\frac{L}{2}-1)\times(\frac{L}{2}-1)$ matrix $\mathbf{A}'$ in the subspace orthogonal to the zero mode is constructed as

$$\mathbf{A}'_{k\ell} = \mathbf{A}_{k\ell} - \mathbf{A}_{k,\frac{L}{2}} - \mathbf{A}_{\frac{L}{2},\ell} + \mathbf{A}_{\frac{L}{2},\frac{L}{2}}, \quad k,\ell = 1,\cdots,\tfrac{L}{2}-1. \tag{3.12}$$

This relation can be easily generalized to higher genus. For a genus-$g$ surface, the discretized expression for the partition function reads

$$Z(L;\ell_a) = \int \prod_{k=1}^{L} dX(k) \prod_{k\in\mathcal{K}} \delta(X(k)-X(k'))\,\Psi_1[X] \propto \frac{1}{\sqrt{\det \mathbf{A}'(L;\ell_a)}}, \tag{3.13}$$

where the matrix $\mathbf{A}(L;\ell_a)$ is defined by (3.9), with the identification map replaced by (2.15). The matrix $\mathbf{A}'(L;\ell_a)$ is obtained from $\mathbf{A}(L;\ell_a)$ by imposing the condition $\sum_{k\in\mathcal{K}} X(k) = 0$ similarly to (3.12).

Let us finish this section with a few remarks. First, the algorithm outlined above can be easily generalized to 1-point functions, where one simply replaces the wavefunctional $\Psi_1[X]$ of the identity operator with the wavefunctional $\Psi_{\mathcal{O}}[X]$ of the corresponding operator. The discretized expression for the 1-point function $\langle\mathcal{O}\rangle$ on the Riemann surface reads

$$\langle\mathcal{O}\rangle_{\Sigma(\ell_a)} = \frac{1}{Z(L;\ell_a)} \int \prod_{k=1}^{L} dX(k) \prod_{k\in\mathcal{K}} \delta(X(k)-X(k'))\,\Psi_{\mathcal{O}}[X]. \tag{3.14}$$

Since the operator $\mathcal{O}$ generally has nonzero conformal dimension, one should be careful about the conformal factor that comes with the local coordinates where the operator is inserted.

The quantity $Z(L;\ell_a)$ computed in this way is subject to a Weyl anomaly, and the corresponding Weyl-anomaly factor generally depends on the moduli $\frac{\ell_a}{L}$, or, equivalently, on the period matrix $\Omega$. In particular, one cannot directly compare the genus-1 partition function $Z(L;\ell_1,\ell_2)$ with the analytic expression obtained in the flat-torus frame, since a nontrivial

Weyl transformation relates the disk and flat-torus frames. However, one can compare the partition functions of two different theories with the same central charge at the same moduli, or compute the 1-point function $\langle \mathcal{O} \rangle_{\Sigma(\ell_a)}$ of a nontrivial operator, and such observables are free of the Weyl-anomaly factor.

The quantity $Z(L; \ell_1, \ell_2)$ is also divergent as a function of the total length $L$ in the continuum limit $L \to \infty$. There are two sources of this divergence. One comes from the discretization scheme and the normalization of the Gaussian measure, and scales like $\exp(\text{const} \cdot L)$. This piece is clearly moduli-independent. The other comes from the divergence of the Weyl-anomaly factor, which scales like a constant power of $L$. This divergent behavior can be estimated directly as follows.

Around a cubic vertex of the ribbon graph, the metric (2.3) has the behavior $ds^2 = e^{2\omega} dz d\bar{z} \sim |z| dz d\bar{z}$, and so the Weyl transformation $\omega$ that relates the disk metric to the locally flat metric behaves like $\omega \sim \frac{1}{2} \log |z|$. Near the puncture, on the other hand, we have $\omega \sim -\log |z|$. The Weyl anomaly factor in going from the locally flat metric to the ribbon graph metric on a genus-$g$ Riemann surface is given by $\exp(S_{\text{Weyl}})$ with

$$S_{\text{Weyl}} = \frac{1}{12\pi} \int_\Sigma d^2 z \, \partial\omega \bar{\partial}\omega \sim \frac{2g-1}{24} \log \frac{1}{\delta} + \frac{1}{12} \log \frac{1}{\epsilon}, \tag{3.15}$$

where $\delta$ and $\epsilon$ are the cutoff parameters near the $(4g-2)$ cubic vertices and the puncture, respectively. From here, one can see that the coefficient of the logarithmic divergence is independent of the moduli $\Omega$.

If we want to have a finite answer when taking the $L \to \infty$ limit, we need to further renormalize the action by adding counterterms. We define the quantity $Z(L; \ell_a)$ to be

$$Z(L; \ell_a) = \frac{1}{\sqrt{\det \mathbf{A}'(L; \ell_a)}}, \tag{3.16}$$

which amounts to absorbing the constant factor in (3.13) into the renormalization of the world-sheet cosmological constant. The large-$L$ behavior of this partition function takes the form

$$\log Z(L; \ell_a) \sim c(\Omega) + \gamma L + \alpha(g) \log L, \tag{3.17}$$

where $c(\Omega)$ is the moduli-dependent finite part. One can compute $c(\Omega)$, $\gamma$, and $\alpha(g)$ by fitting the large-$L$ behavior of $Z(L; \ell_a)$ with the relative lengths $\frac{\ell_a}{L}$ held fixed. As argued above, $\gamma$ and $\alpha$ are moduli-independent coefficients. Moreover, as the behavior (3.15) suggests, the coefficient $\alpha(g)$ depends linearly on the genus $g$, and one can write

$$\alpha(g) = \alpha_0 + \alpha_1 g. \tag{3.18}$$

In order to arrive at a finite partition function in the $L \to \infty$ limit, we choose the renormalization scheme by adding the following counterterms to the free boson action:

$$S_{\text{CT}} = \gamma L + (\alpha_0 + \alpha_1) \log L - \frac{\alpha_1}{8\pi} \log L \int_\Sigma \mathrm{d}^2\sigma \sqrt{g} \, R(g). \tag{3.19}$$

These counterterms will cancel the divergences in (3.17), and the renormalized partition function in this scheme is simply given by

$$Z_{\text{ren}}(\Omega) = \exp c(\Omega). \tag{3.20}$$

Of course, the renormalization scheme here is not unique, and one can perform a finite renormalization, namely adding an arbitrary finite part to the counterterms, to define a different renormalization scheme.

## 3.2 The compact boson partition function

The algorithm above generalizes to the compact boson CFT. Consider the genus-$g$ partition function of a compact boson $X$ satisfying $X \sim X + 2\pi R$. On the unit circle, a configuration of $X$ can be parametrized by the winding number $p \in \mathbb{Z}$ and the Fourier modes $X_m$:

$$X_b(\sigma) = p\sigma R + \sum_{m=-\infty}^{\infty} X_m e^{im\sigma}. \tag{3.21}$$

For any configuration on the disk with nonzero winding number on the boundary, the classical action is logarithmically divergent. Consequently, the wavefunctional $\Psi_1[X_b] = \int [\mathcal{D}X]_{X_b} e^{-S[X]}$ vanishes. Therefore, $\Psi_1[X_b]$ is identical to the wavefunctional of a non-compact boson in the zero-winding sector and vanishes in sectors with nontrivial winding:

$$\Psi_1[p, X_m] \propto \delta_{p,0} \exp\left[-\frac{1}{\alpha'} \sum_{m\geq 1} m X_m X_{-m}\right]. \tag{3.22}$$

This wavefunctional can then be discretized as in (3.5).

For the compact boson, the gluing map of boundary fields $X(k)$ should be modified to include an integer shift $s_a \in \mathbb{Z}$ for each edge $E_a$. The most general gluing map reads

$$X\left(\sum_{j\leq i_1(a)} \ell_{a(j)} + \sum_{k<i_2(a)} \ell_{a(k)} - k + 1\right) = X(k) + 2\pi R s_a, \quad \sum_{j<i_1(a)} \ell_{a(j)} + 1 \leq k \leq \sum_{j\leq i_1(a)} \ell_{a(j)}, \tag{3.23}$$

which includes $6g-3$ integer shifts. However, on a genus-$g$ surface, there are only $2g$ independent 1-cycles, which means that there should be $4g-3$ linear relations among the $6g-3$ shifts. These relations come from the fact that a field configuration with finite action should have no winding around each contractible cycle. Specifically, the three shifts associated with going around each Strebel vertex should add up to zero, as illustrated in figure 5.

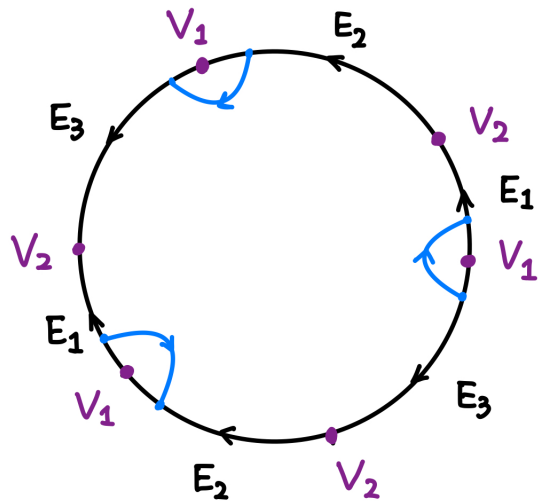


Figure 5: The no-winding condition $s_1 - s_2 + s_3 = 0$ around the Strebel vertex $V_1$ for a genus-1 Riemann surface. The contractible cycle around $V_1$ is shown in blue.

For each Strebel vertex $V_A$ ($A = 1, \cdots, 4g-2$), this condition can be written as

$$\sum_{a=1}^{6g-3} D_{A,a} s_a = 0, \tag{3.24}$$

where $D_{A,a}$ is defined as follows. For the ribbon graph $\Gamma$ that characterizes the Riemann surface $\Sigma$, one can define an oriented graph $\hat{\Gamma}$ as follows: the edges and vertices of $\hat{\Gamma}$ are the same as those of $\Gamma$, and the orientation of edge $E_a$ in $\hat{\Gamma}$ is taken to be the counter-clockwise direction of the boundary segment $S_{i_1(a)}$ of the disk, where $S_{i_1(a)}$ is the first boundary segment associated with the edge $E_a$. The matrix $D_{A,a}$ is then given by the incidence matrix of the oriented graph $\hat{\Gamma}$:

$$D_{A,a} = \begin{cases} 1 & \text{if edge } E_a \text{ comes into vertex } V_A, \\ -1 & \text{if edge } E_a \text{ goes out of vertex } V_A, \\ 0 & \text{if edge } E_a \text{ is not connected to vertex } V_A. \end{cases} \tag{3.25}$$

The matrix equation (3.24) then tells us that the independent shifts live in the kernel $\ker_{\mathbb{Z}}(D)$, which is a $2g$-dimensional lattice that corresponds exactly to the shifts around the $2g$ independent 1-cycles on the Riemann surface $\Sigma$.

A basis of $\ker_{\mathbb{Z}}(D)$ can be built in the following way. One first chooses a spanning tree of the oriented graph $\hat{\Gamma}$, which contains $4g-3$ edges. For each of the $2g$ edges $E_e$ that are not in the spanning tree, we denote its start point and end point by $V_{A_1(e)}$ and $V_{A_2(e)}$, respectively. Given such a non-tree edge $E_e$, one can always find a cycle $C_e$ in $\hat{\Gamma}$ by starting from $V_{A_1(e)}$,

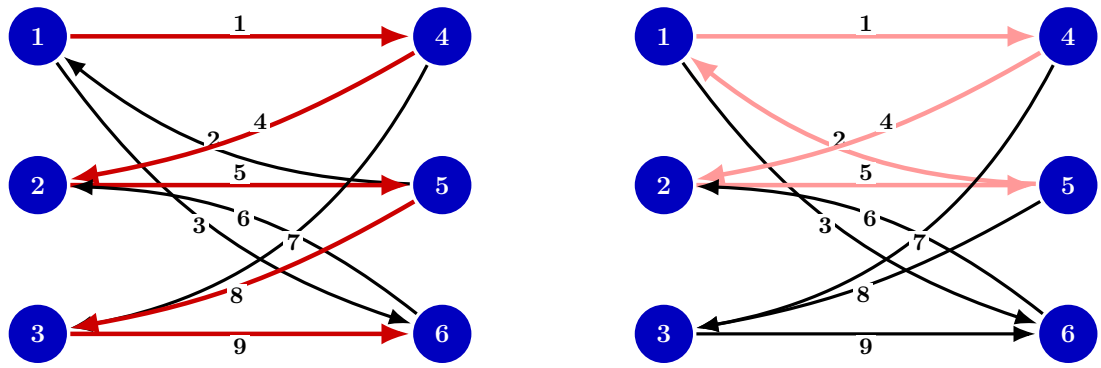


Figure 6: Left: The oriented graph $\hat{\Gamma}$ associated with a genus-2 ribbon graph $\Gamma$, with a spanning tree highlighted in red. Right: The cycle $C_e$ in $\hat{\Gamma}$ for the non-tree edge $e = 2$, shown in pink.

going through $E_e$ to get to $V_{A_2(e)}$, and then taking the unique path in the spanning tree from $V_{A_2(e)}$ back to $V_{A_1(e)}$, as shown in figure 6. From this cycle $C_e$, one then defines a lattice vector $b_e \in \mathbb{Z}^{6g-3}$ by

$$(b_e)_a = \begin{cases} 1 & \text{if edge } a \text{ is traversed with the same orientation in } C_e, \\ -1 & \text{if edge } a \text{ is traversed with the opposite orientation in } C_e, \\ 0 & \text{if edge } a \text{ is not in } C_e, \end{cases} \tag{3.26}$$

and the $2g$ vectors $\{b_e\}$ will form an integral basis of $\ker_{\mathbb{Z}}(D)$. If we define the matrix $\mathbf{B} \in \mathbb{Z}^{(6g-3)\times 2g}$ by

$$\mathbf{B}_{a,e} = (b_e)_a, \tag{3.27}$$

then an arbitrary lattice vector $s$ in $\ker_{\mathbb{Z}}(D)$ can be represented by

$$s = \mathbf{B}n, \tag{3.28}$$

where $n \in \mathbb{Z}^{2g}$ is a lattice vector in the $2g$-dimensional integral lattice. Namely, the physical shifts $s_a$ all take the form (3.28).

The discretized compact boson partition function is then obtained by summing over all physical shifts

$$Z(R; L; \ell_a) = \sum_{n\in\mathbb{Z}^{2g}} \int \prod_{k=1}^{L} dX(k) \prod_{a=1}^{6g-3} \prod_{k\in S_{i_1(a)}} \delta\left(X(k') - X(k) - 2\pi R s_a(n)\right) \Psi_1[X], \tag{3.29}$$

where $s_a(n)$ is given by (3.28) in terms of the lattice vector $n$. After replacing the variables $X(k)$ with $k \in \{1, \cdots, L\}\backslash\mathcal{K}$ using the delta functions and factoring out the zero mode, one

obtains

$$Z(R;L;\ell_a) = 2\pi R \sum_{n\in\mathbb{Z}^{2g}} \int_{k\in\mathcal{K}'} \prod_{k\in\mathcal{K}} dX(k) \exp\left[-\pi^{-1}\left(\sum_{k,\ell\in\mathcal{K}'} \mathbf{A}'_{k\ell} X(k)X(\ell) + 2\sum_{k\in\mathcal{K}'} \mathbf{v}'_k X(k) + \mathbf{C}\right)\right], \tag{3.30}$$

where the set $\mathcal{K}'$ is defined to be $\mathcal{K}\backslash\{1\}$. The matrix $\mathbf{A}'$, the vector $\mathbf{v}'$, and the constant $\mathbf{C}$ are defined as

$$\begin{aligned}
\mathbf{M}_{k\ell} &= \frac{1}{2\alpha' L} \frac{\sin\frac{\pi}{L}}{\cos\frac{2\pi(k-\ell)}{L} - \cos\frac{\pi}{L}}, \\
\mathbf{A}_{k\ell} &= \mathbf{M}_{k\ell} + \mathbf{M}_{k'\ell} + \mathbf{M}_{k\ell'} + \mathbf{M}_{k'\ell'}, \quad \mathbf{A}'_{k\ell} = \mathbf{A}_{k\ell} - \mathbf{A}_{1,\ell} - \mathbf{A}_{k,1} + \mathbf{A}_{1,1}, \\
\mathbf{v}_k &= 2\pi R \sum_{a=1}^{6g-3} s_a(n) \sum_{\ell\in S_{i_1(a)}} (\mathbf{M}_{k\ell'} + \mathbf{M}_{k'\ell'}), \quad \mathbf{v}'_k = \mathbf{v}_k - \mathbf{v}_1, \\
\mathbf{C} &= (2\pi R)^2 \sum_{a,b=1}^{6g-3} s_a(n) s_b(n) \sum_{k\in S_{i_1(a)}} \sum_{\ell\in S_{i_1(b)}} \mathbf{M}_{k'\ell'}.
\end{aligned} \tag{3.31}$$

Performing the Gaussian integral then gives

$$Z(R;L;\ell_a) \propto \frac{R}{\sqrt{\det \mathbf{A}'}} \sum_{n\in\mathbb{Z}^{2g}} \exp\left(-4\pi R^2 n^{\mathrm{T}} \mathbf{T}' n\right) = \frac{R}{\sqrt{\det \mathbf{A}'}} \Theta(4iR^2\mathbf{T}'), \tag{3.32}$$

where $\Theta$ is the Siegel theta function, and the $2g\times 2g$ matrix $\mathbf{T}'$ is given by

$$\begin{aligned}
\mathbf{W}_{k\ell} &= \mathbf{M}_{k\ell'} + \mathbf{M}_{k'\ell'} - \mathbf{M}_{1,\ell'} - \mathbf{M}_{1',\ell'}, \\
\mathbf{T}_{ab} &= \sum_{k\in S_{i_1(a)}} \sum_{\ell\in S_{i_1(b)}} \left(\mathbf{M}_{k'\ell'} - \sum_{m,n\in\mathcal{K}'} (\mathbf{A}')^{-1}_{mn} \mathbf{W}_{mk} \mathbf{W}_{n\ell}\right), \\
\mathbf{T}' &= \mathbf{B}^{\mathrm{T}} \mathbf{T} \mathbf{B},
\end{aligned} \tag{3.33}$$

and the matrix $\mathbf{B}$ is defined in (3.27).

# 4 $bc$ ghosts and string amplitudes

Correlators of the fermionic $(b_\lambda, c_{1-\lambda})$ system on higher-genus Riemann surfaces can be computed from the holomorphic data obtained in section 2.3 using the formula [5]

$$\begin{aligned}
\left\langle \prod_{i=1}^{n} b_\lambda(z_i) \prod_{j=1}^{m} c_{1-\lambda}(w_j) \right\rangle_\Sigma &= Z_{\text{chiral}}\, \theta\left(\sum_i \zeta(z_i) - \sum_j \zeta(w_j) - (2\lambda-1)\Delta \mid \Omega\right) \\
&\quad \times \frac{\prod_{i<i'} E(z_i, z_{i'}) \prod_{j<j'} E(w_j, w_{j'}) \prod_i (\sigma(z_i))^{2\lambda-1}}{\prod_{i,j} E(z_i, w_j) \prod_j (\sigma(w_j))^{2\lambda-1}},
\end{aligned} \tag{4.1}$$

where $n$ and $m$ must saturate the ghost number anomaly $n - m = (2\lambda - 1)(g - 1)$. Below, we explain the chiral boson partition function $Z_{\text{chiral}}$, the Abel-Jacobi map $\zeta(z)$, the Riemann theta function $\theta(y|\Omega)$, the prime form $E(z, w)$, the Riemann constant vector $\Delta$, and the function $\sigma(z)$. We also describe the algorithm used to compute these quantities from the holomorphic 1-forms $\omega_I$ and the period matrix $\Omega$ obtained in section 2.3.

### The chiral boson partition function

The chiral boson partition function $Z_{\text{chiral}}$ is the holomorphic conformal block of the $U(1)$ current algebra generated by $\partial X$ in the free boson CFT. It can be related to the free boson partition function by

$$Z_{\text{boson}}(\Omega) = V_X \int \frac{d^g k}{(2\pi)^g} e^{-\pi k^I k^J \text{Im}(\Omega_{IJ})} |Z_{\text{chiral}}|^2 \propto [\det \text{Im}(\Omega)]^{-1/2} |Z_{\text{chiral}}|^2. \tag{4.2}$$

Both $Z_{\text{boson}}$ and $Z_{\text{chiral}}$ depend on the renormalization scheme. Moreover, the quantity $Z_{\text{chiral}}$ depends on the Weyl frame and is subject to a gravitational anomaly due to the unbalanced central charge $(1, 0)$. In principle, to specify $Z_{\text{chiral}}$ fully, one must choose a Hermitian metric on the worldsheet $\Sigma$ as well as a conformal frame that specifies the plumbing fixtures used to construct $\Sigma$ by sewing along the $\alpha^I$ cycles. The quantity $|Z_{\text{chiral}}|^2$ depends only on the Weyl frame, and we take that Weyl frame to be the same one used for free-boson correlators in bosonic string amplitudes.

In practice, we can choose the worldsheet metric where $Z_{\text{chiral}}$ is given by

$$Z_{\text{chiral}} = \left[ \frac{\det \text{Im}(\Omega)}{(\det \mathbf{A}'(\Omega))_{\text{ren}}} \right]^{\frac{1}{4}}, \tag{4.3}$$

where the branch of the fourth root is chosen such that $Z_{\text{chiral}}$ is always positive, and we choose the renormalization scheme such that the renormalized determinant $(\det \mathbf{A}'(\Omega))_{\text{ren}}$ is defined similarly to (3.20), namely $\log (\det \mathbf{A}'(\Omega))_{\text{ren}}$ is taken to be the finite part of $\log \det \mathbf{A}'(L; \ell_a)$ after stripping off the large-$L$ divergence of the form $\gamma L + \alpha \log L$. For physical amplitudes, all dependence on the worldsheet metric will cancel out since such amplitudes only depend on $|Z_{\text{chiral}}|^2$. The Weyl frame of (4.3) will automatically agree with that of the free boson correlators computed from section 3.

### The Abel-Jacobi map

The Abel-Jacobi map $\zeta(z) = (\zeta_1(z), ..., \zeta_g(z))$ is defined as

$$\zeta_I(z) = \int_{z_0}^{z} \omega_I, \tag{4.4}$$

where $z_0$ is a fixed point on $\Sigma$ and $\omega_I$ are the holomorphic 1-forms. Note that (4.4) depends on the integration contour from $z_0$ to $z$, and $\zeta(z)$ is defined modulo shifts by periods

$$\zeta_I(z) \to \zeta_I(z) + m_I + \Omega_{IJ} n^J, \tag{4.5}$$

where $m_I$ and $n^I$ are lattice vectors in $\mathbb{Z}^g$. In practice, we compute $\zeta(z)$ using (4.4) by taking $z_0$ to be the origin of the unit disk and the contour to be the radial ray from 0 to the point $z$ on the disk.

### Spin structures and the Riemann theta function

The Riemann theta function $\theta(y|\Omega)$ that enters (4.1) directly is defined as

$$\theta(y|\Omega) = \sum_{n\in\mathbb{Z}^g} \exp\left(\pi i n^I \Omega_{IJ} n^J + 2\pi i n^I y_I\right). \tag{4.6}$$

Given a spin structure on $\Sigma$, one can generalize this definition as follows. A spin structure on $\Sigma$ can be parametrized by the characteristic

$$\delta = (\delta^I, \delta'_I) \in \left\{0, \tfrac{1}{2}\right\}^{2g}, \tag{4.7}$$

where the periodicity of a spinor field along the $\alpha^I$ cycle is given by $\epsilon^I = (-1)^{2\delta^I - 1}$ in the conformal frame in which the cycle $\alpha^I$ can be represented by the circumference of a flat cylinder. The spinor periodicity along $\beta_I$ is given by $\epsilon'_I = (-1)^{2\delta'_I - 1}$ in a similar conformal frame. The Arf invariant of the spin structure $\delta$ is given by

$$\mathrm{Arf}(\delta) = 4\sum_I \delta^I \delta'_I \pmod 2. \tag{4.8}$$

It's clear that $\mathrm{Arf}(\delta)$ can only be 0 or 1. The spin structures $\delta$ with $\mathrm{Arf}(\delta) = 0$ are called even, and those with $\mathrm{Arf}(\delta) = 1$ are called odd. The Riemann theta function with characteristic is then defined by

$$\theta[\delta](y|\Omega) = \sum_{n\in\mathbb{Z}^g} \exp\left[\pi i (n^I + \delta^I)\Omega_{IJ}(n^J + \delta^J) + 2\pi i (n^I + \delta^I)(y_I + \delta'_I)\right]. \tag{4.9}$$

### The prime form

The prime form $E(z,w)$ is the unique weight-$(-\frac{1}{2})$ holomorphic differential in both $z$ and $w$ that vanishes only at $z = w$. It can be computed explicitly using

$$E(z,w) = \frac{\theta[\delta](\zeta(z) - \zeta(w)|\Omega)}{\sqrt{\omega[\delta](z)\omega[\delta](w)}}, \tag{4.10}$$

where $\delta$ parametrizes an odd spin structure on $\Sigma$. It can be proved that (4.10) is independent of the odd spin structure $\delta$. The quantity $\omega[\delta](z)$ is a special holomorphic 1-form defined as

$$\omega[\delta](z) = \omega_I(z)\frac{\partial}{\partial y_I}\theta[\delta](y|\Omega)|_{y=0}. \tag{4.11}$$

In practice, the Riemann theta function is computed by truncating on the lattice sum, and the prime form follows directly from definition (4.10) and (4.11).

### The Riemann constant vector

The Riemann constant vector $\Delta = (\Delta_1, \cdots, \Delta_g)$ is the vector that appears in the Riemann vanishing theorem, whose statement is that $\theta(y|\Omega)$ vanishes if and only if $y$ can be written as

$$y = \sum_{i=1}^{g-1} \zeta(z_i) - \Delta \tag{4.12}$$

where $\{z_i\}$ are a set of $g-1$ points on the Riemann surface $\Sigma$. We adopt the algorithm in [6] for computing $\Delta$, which is based on the following theorem [7].

**Theorem 2** *Let $\omega$ be a nonzero holomorphic 1-form on the Riemann surface $\Sigma$ with distinct zeros $\{c_r\}$, where the zero at $c_r$ has degree $m_r$. Then the Riemann constant vector $\Delta$ satisfies*

$$2\Delta = \sum_r m_r \zeta(c_r) \tag{4.13}$$

*modulo the period ambiguity* (4.5).

In the code, we take the holomorphic 1-form to be the first $\alpha$-normalized holomorphic 1-form $\hat{\omega}_1$, defined as

$$\hat{\omega}_I = (\mathcal{A}^{-1})_I{}^J \omega_J, \tag{4.14}$$

where the matrix $\mathcal{A}$ is defined in (2.21). The zeros of $\hat{\omega}_1$ are obtained as follows. One first computes the coefficients $\{a_n\}_{n=1}^{L/2}$ for $\hat{\omega}_1$ in (2.17) from the known coefficients of $\omega_I$ and the matrix $\mathcal{A}$. The zeros $c_r$ of $\hat{\omega}_1$ are the roots of the polynomial $\sum_{n=1}^{L/2} a_n z^n$, with the same degrees. After obtaining the zeros and their degrees, we construct the candidate

$$\Delta_0 \equiv \frac{1}{2}\sum_{\text{zeros } r} m_r \zeta(c_r). \tag{4.15}$$

Since the Abel-Jacobi map $\zeta(z)$ is subject to the ambiguity (4.5), the equation (4.13) only determines $\Delta$ up to a half-period, so one enumerates the $2^{2g}$ candidates

$$\Delta_{\text{cand}} = \Delta_0 + m + \Omega n, \qquad m, n \in \{0, \tfrac{1}{2}\}^g. \tag{4.16}$$

To select the correct $\Delta$ among these candidates, we test the Riemann vanishing theorem on a small fixed set of points $\{p_k\}$ in the interior of the disk, and choose the candidate $\Delta_{\text{cand}}$ that minimizes

$$\sum_k \left|\theta\Big((g-1)\zeta(p_k) - \Delta_{\text{cand}} \,\Big|\, \Omega\Big)\right|^2 . \tag{4.17}$$

**The function $\sigma(z)$**

# 5 Comparison with analytic results

## 5.1 The stress-tensor torus 1-point function

## 5.2 Compact boson partition functions

## 5.3 T-duality

## 5.4 The Igusa cusp form

## 5.5 The genus-3 Mumford form

# 6 Discussions

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